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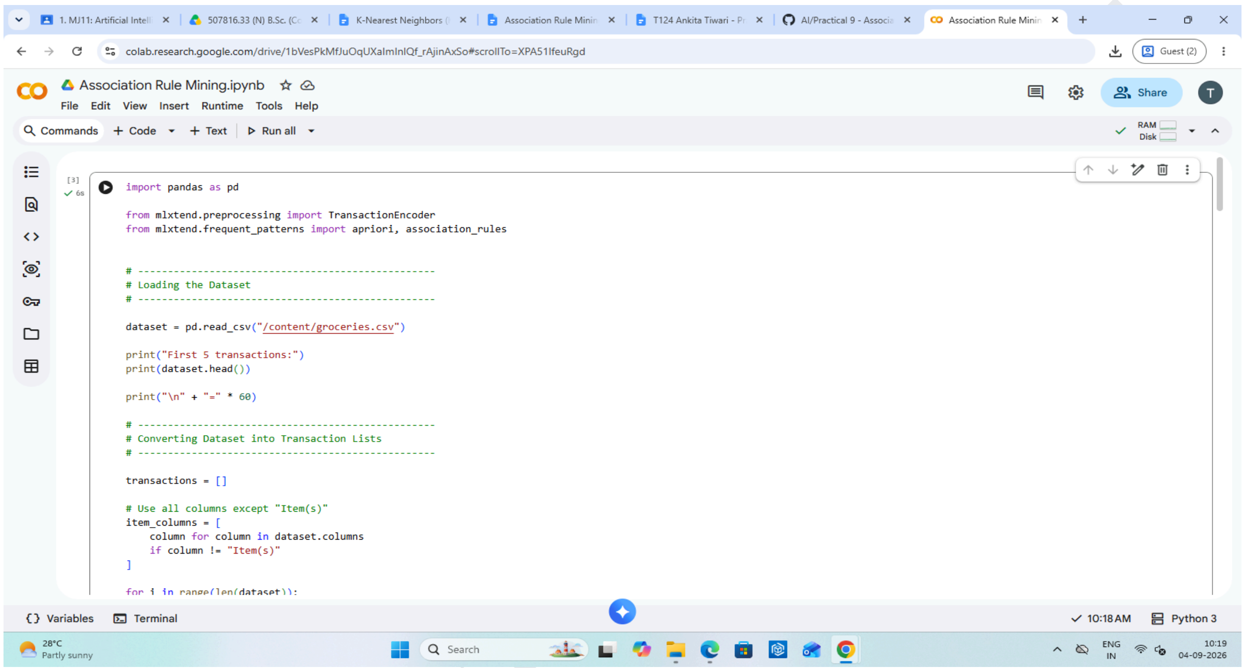
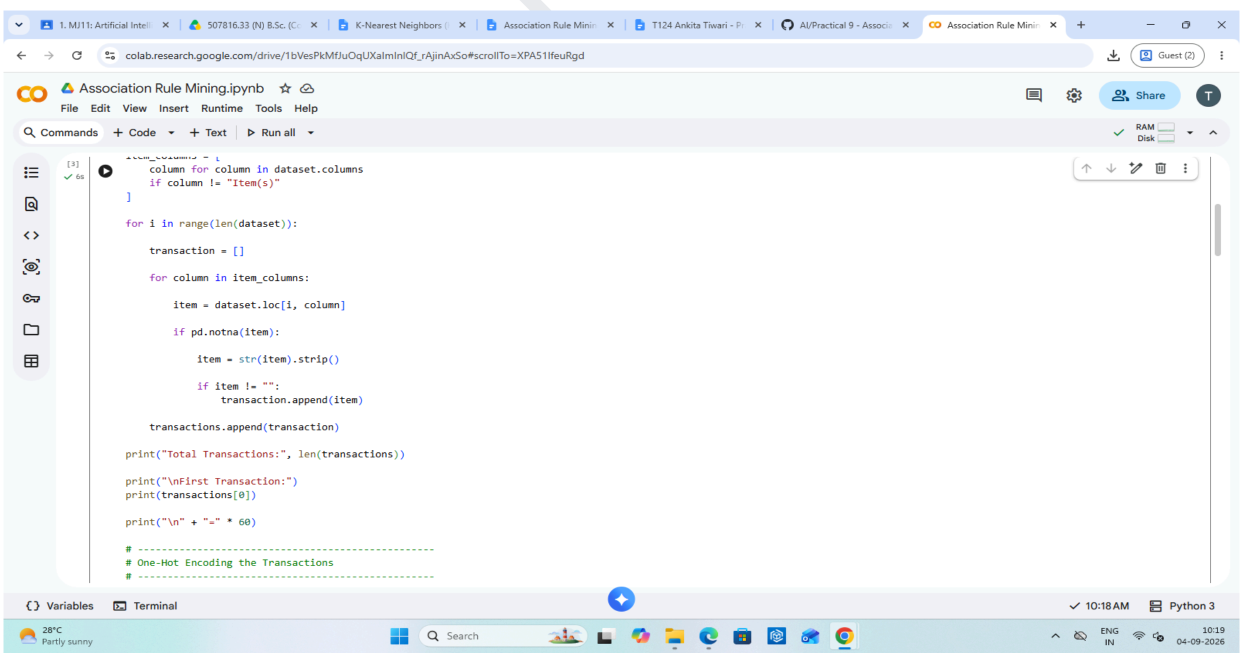
AIM:Association Rule Mining
Implement the Association Rule Mining algorithm (e.g., Apriori) to find frequent itemsets.
Generate association rules from the frequent itemsets and calculate their support and confidence.
Interpret and analyze the discovered association rules.
INPUT:



```
import pandas as pd

from mlxtend.preprocessing import TransactionEncoder
from mlxtend.frequent_patterns import apriori, association_rules


# --------------------------------------------------
# Loading the Dataset
# --------------------------------------------------

dataset = pd.read_csv("/content/groceries.csv")

print("First 5 transactions:")
print(dataset.head())

print("\n" + "=" * 60)

# --------------------------------------------------
# Converting Dataset into Transaction Lists
# --------------------------------------------------

transactions = []

# Use all columns except "Item(s)"
item_columns = [
    column for column in dataset.columns
    if column != "Item(s)"
]

for i in range(len(dataset)):
```



```
    column for column in dataset.columns
    if column != "Item(s)"
]

for i in range(len(dataset)):

    transaction = []

    for column in item_columns:

        item = dataset.loc[i, column]

        if pd.notna(item):

            item = str(item).strip()

            if item != "":
                transaction.append(item)

    transactions.append(transaction)

print("Total Transactions:", len(transactions))

print("\nFirst Transaction:")
print(transactions[0])

print("\n" + "=" * 60)

# --------------------------------------------------
# One-Hot Encoding the Transactions
# --------------------------------------------------
```

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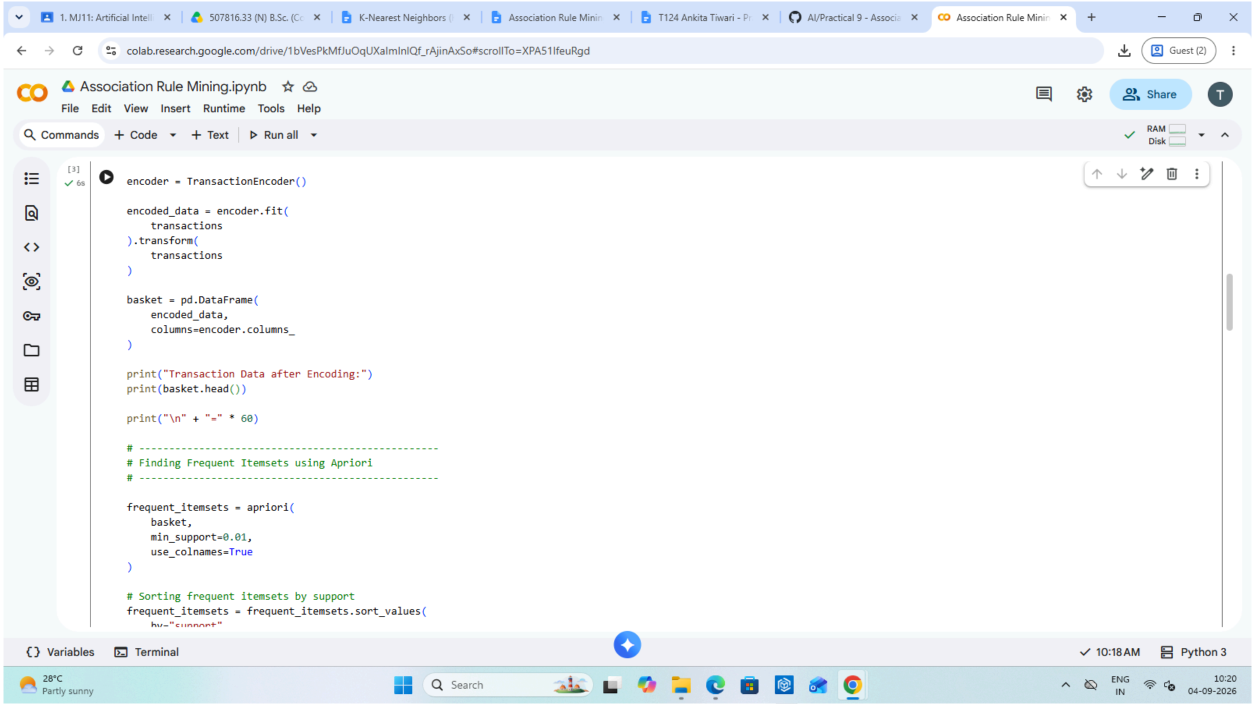
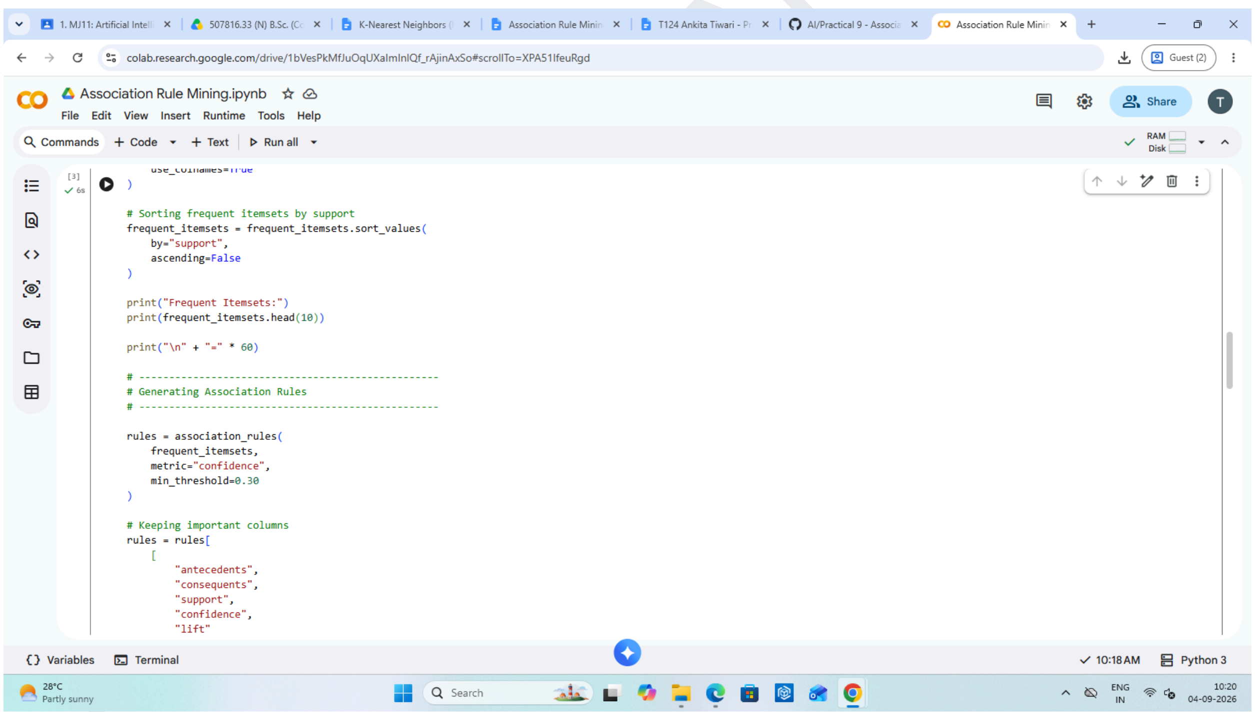


```
encoder = TransactionEncoder()

encoded_data = encoder.fit(
    transactions
).transform(
    transactions
)

basket = pd.DataFrame(
    encoded_data,
    columns=encoder.columns_
)

print("Transaction Data after Encoding:")
print(basket.head())

print("\n" + "=" * 60)

# --------------------------------------------------
# Finding Frequent Itemsets using Apriori
# --------------------------------------------------

frequent_itemsets = apriori(
    basket,
    min_support=0.01,
    use_colnames=True
)

# Sorting frequent itemsets by support
frequent_itemsets = frequent_itemsets.sort_values(
```



```
)

# Sorting frequent itemsets by support
frequent_itemsets = frequent_itemsets.sort_values(
    by="support",
    ascending=False
)

print("Frequent Itemsets:")
print(frequent_itemsets.head(10))

print("\n" + "=" * 60)

# --------------------------------------------------
# Generating Association Rules
# --------------------------------------------------

rules = association_rules(
    frequent_itemsets,
    metric="confidence",
    min_threshold=0.30
)

# Keeping important columns
rules = rules[
    [
        "antecedents",
        "consequents",
        "support",
        "confidence",
        "lift"
```

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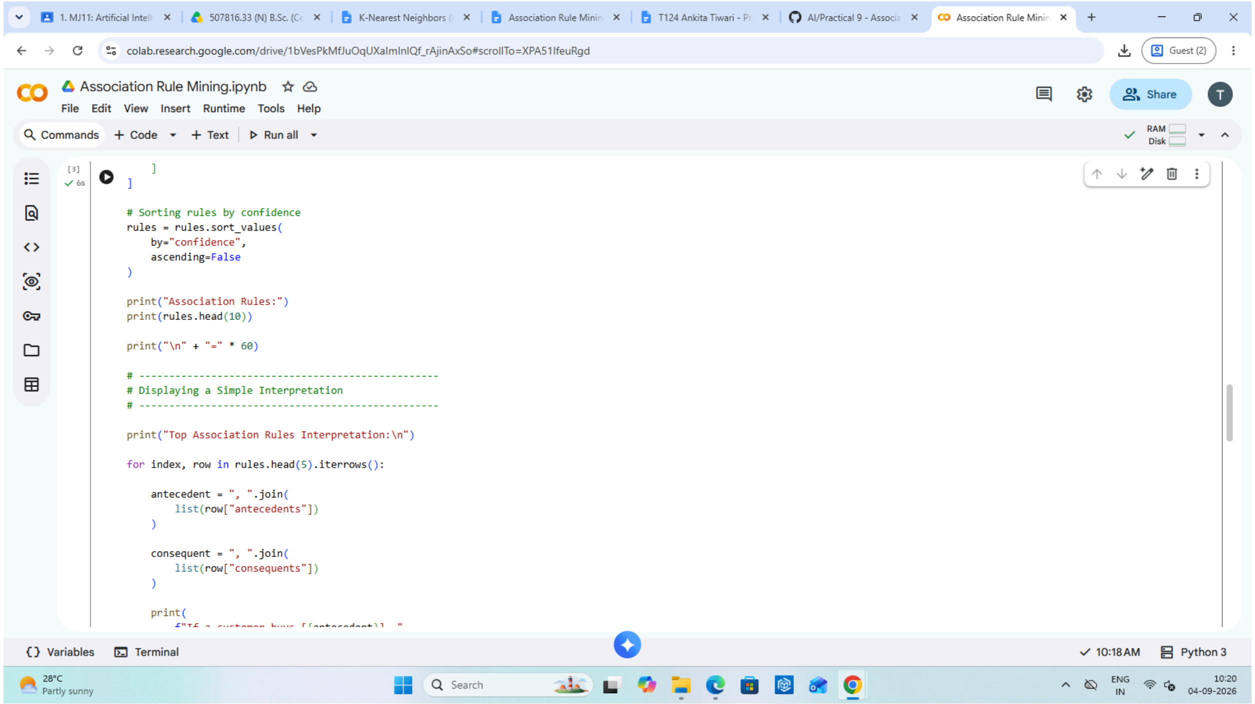

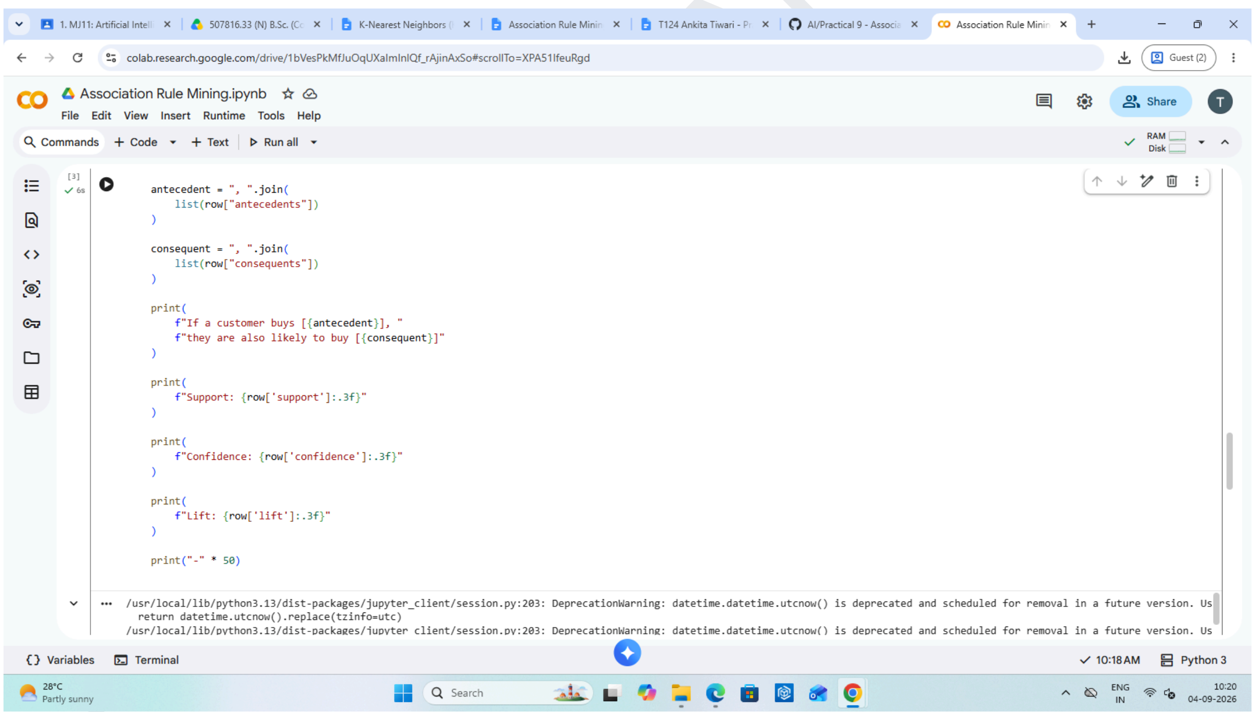

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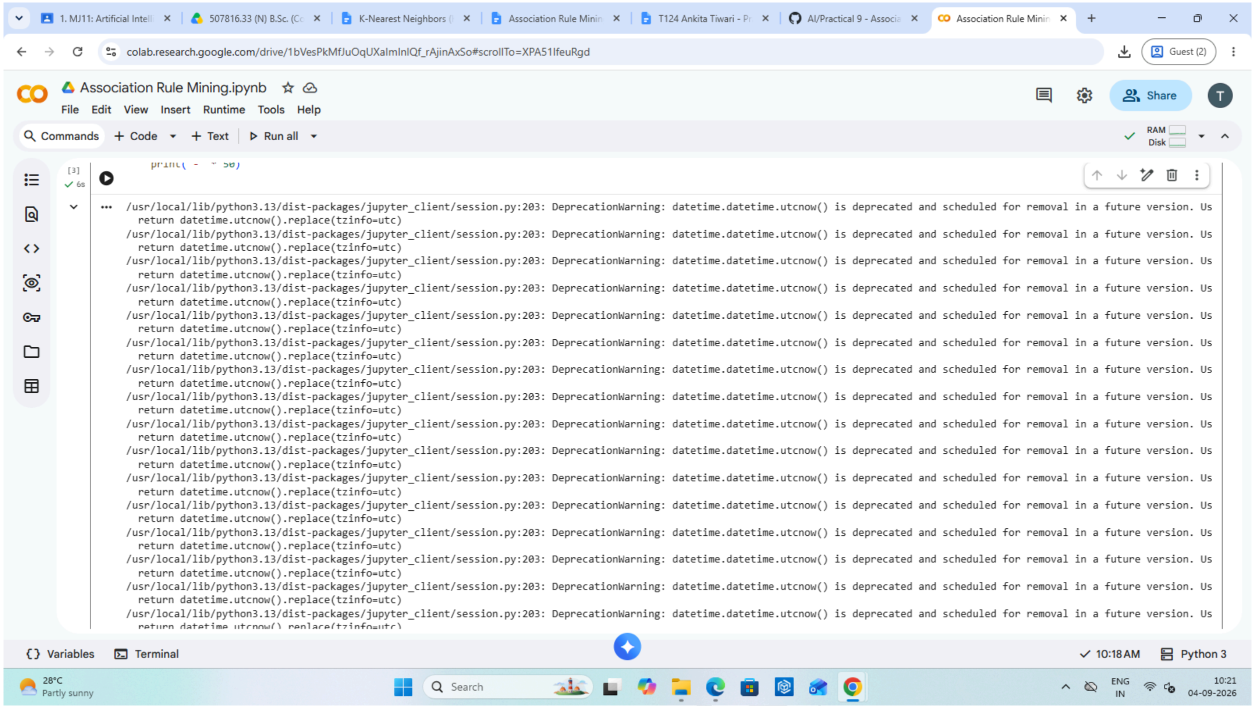
OUTPUT:



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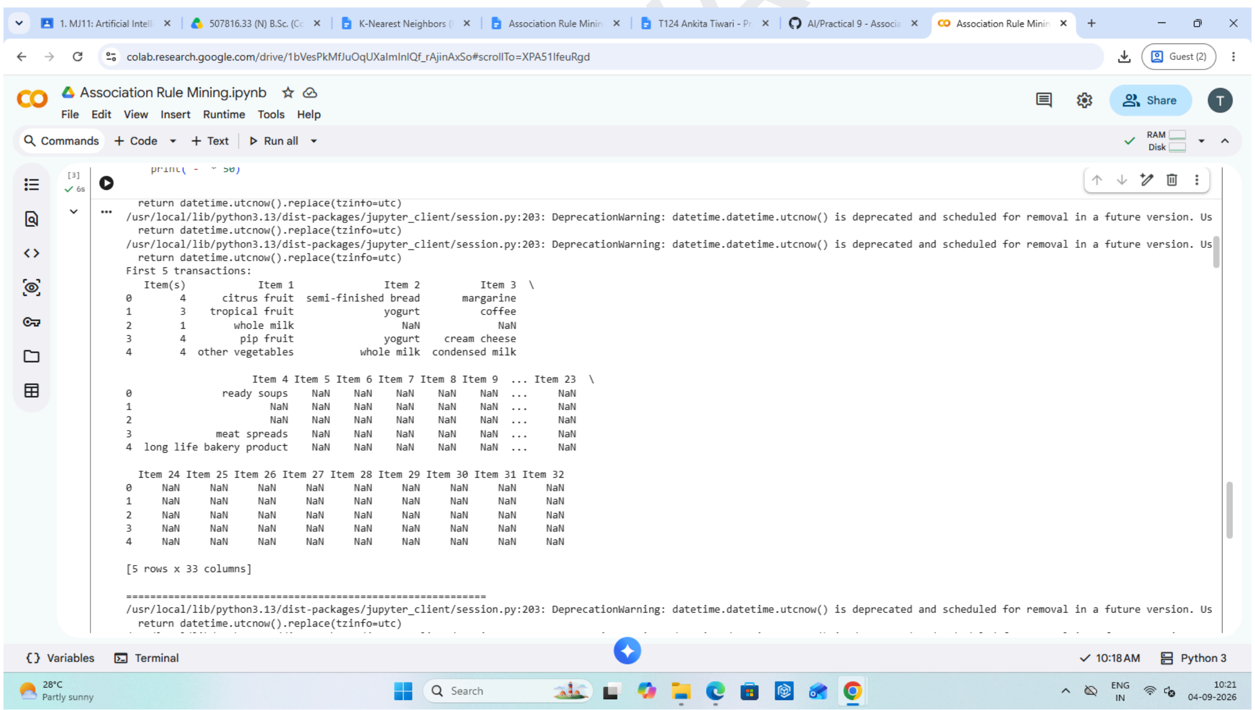

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First 5 transactions:
   Item(s)             Item 1               Item 2           Item 3  \
0        4       citrus fruit  semi-finished bread        margarine
1        3     tropical fruit               yogurt           coffee
2        1         whole milk                  NaN              NaN
3        4          pip fruit               yogurt     cream cheese
4        4   other vegetables           whole milk  condensed milk

                     Item 4 Item 5 Item 6 Item 7 Item 8 Item 9  ... Item 23  \
0               ready soups    NaN    NaN    NaN    NaN    NaN  ...     NaN
1                       NaN    NaN    NaN    NaN    NaN    NaN  ...     NaN
2                       NaN    NaN    NaN    NaN    NaN    NaN  ...     NaN
3              meat spreads    NaN    NaN    NaN    NaN    NaN  ...     NaN
4  long life bakery product    NaN    NaN    NaN    NaN    NaN  ...     NaN

  Item 24 Item 25 Item 26 Item 27 Item 28 Item 29 Item 30 Item 31 Item 32
0     NaN     NaN     NaN     NaN     NaN     NaN     NaN     NaN     NaN
1     NaN     NaN     NaN     NaN     NaN     NaN     NaN     NaN     NaN
2     NaN     NaN     NaN     NaN     NaN     NaN     NaN     NaN     NaN
3     NaN     NaN     NaN     NaN     NaN     NaN     NaN     NaN     NaN
4     NaN     NaN     NaN     NaN     NaN     NaN     NaN     NaN     NaN

[5 rows x 33 columns]

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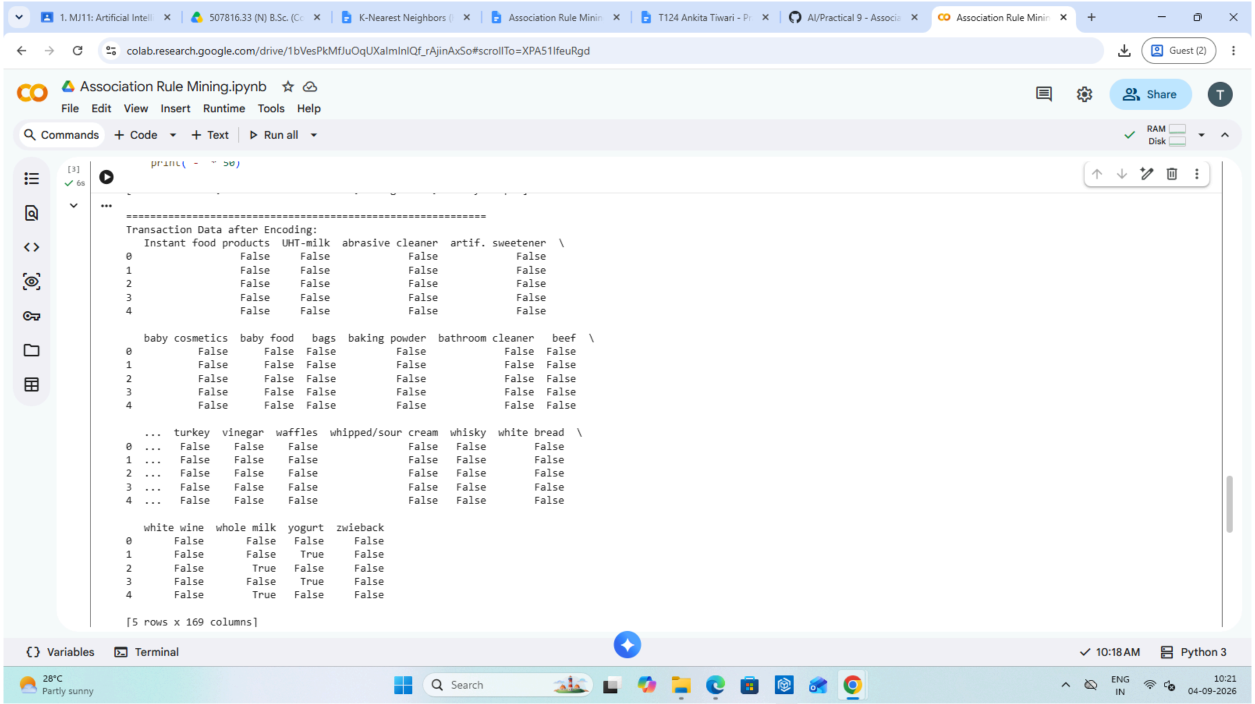

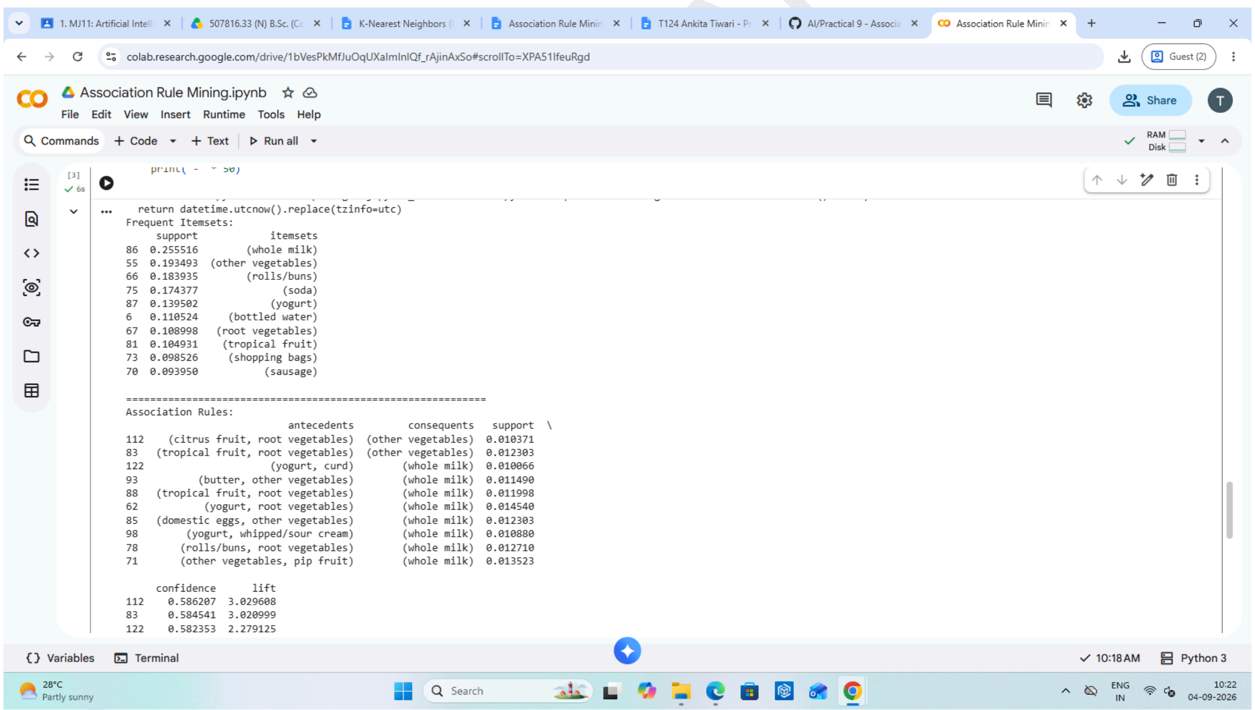

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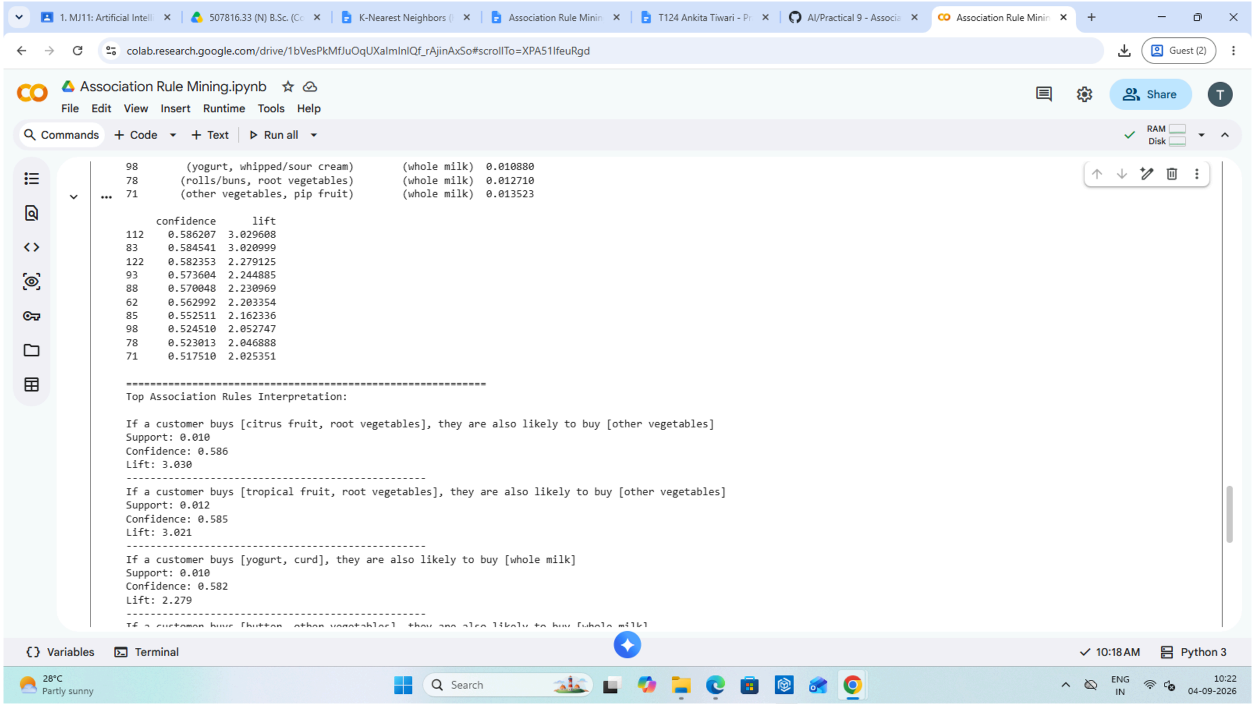

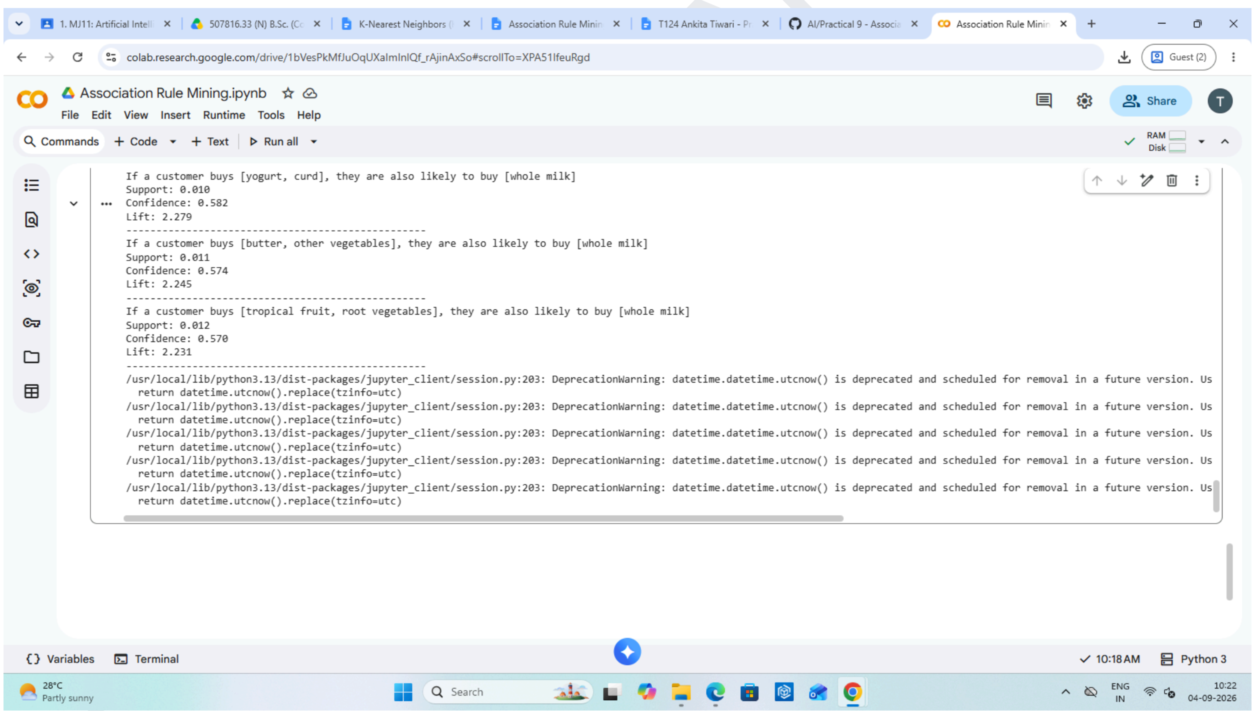

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